INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

🌲 Data Structures & Basic Algorithms Interview Guide

Section: 10_Interview_Preparation | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

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1. Foundational Data Structures

- **Arrays vs Linked Lists:** Arrays provide $O(1)$ random access by index but have fixed contiguous memory; Linked Lists allow dynamic resizing and $O(1)$ insertion/deletion at known nodes but require $O(N)$ traversal.
- **Stack:** LIFO (Last-In-First-Out) used in function recursion call stacks, undo mechanisms, and expression syntax parsing.
- **Queue:** FIFO (First-In-First-Out) used in CPU scheduling, print buffers, and BFS graph traversals.
- **Binary Search Tree (BST):** Binary tree where left subtree nodes $<$ root $<$ right subtree nodes. Search, insertion, and deletion are $O(\log N)$ average, $O(N)$ worst case (skewed tree).

2. Sorting & Searching Complexities

Algorithm	Best Time	Average Time	Worst Time	Space Complexity	Stable?
Binary Search	$O(1)$	$O(\log N)$	$O(\log N)$	$O(1)$	N/A
Bubble Sort	$O(N)$	$O(N^2)$	$O(N^2)$	$O(1)$	Yes
Merge Sort	$O(N \log N)$	$O(N \log N)$	$O(N \log N)$	$O(N)$	Yes
Quick Sort	$O(N \log N)$	$O(N \log N)$	$O(N^2)$	$O(\log N)$	No